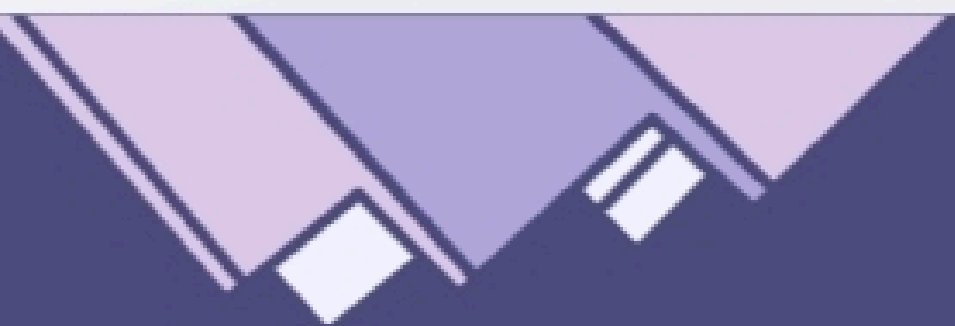
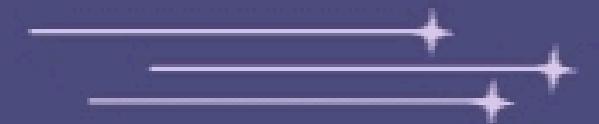




2.2 Complex Numbers

Rectangular Form • Argand Plane • Algebraic Operations

Extending the Real Number System



2.2 Complex Numbers

Why do we need complex numbers?

- The equation $x^2 + 1 = 0$ has no solution in the real number system.
- In general, there are polynomial equations with real coefficients that have no real solution.
- We enlarge the real number system to accommodate solutions of such polynomial equations.
- The imaginary unit i has the property $i^2 = -1$.
- Combining i with two real numbers x and y (by addition and multiplication) gives a complex number $x + iy$. The symbol '+' is treated as vector addition.
- Introduced by Carl Friedrich Gauss (1777 – 1855).

In this section, we define:

- (i) Complex numbers in rectangular form (ii) Argand plane (iii) Algebraic operations

2.2.1 Rectangular Form

Definition of a complex number in rectangular form

Definition 2.1 (Rectangular form of a complex number)

A complex number is of the form $x + iy$ (or $x + yi$), where x and y are real numbers.

x is called the real part and y is called the imaginary part of the complex number.

- If $x = 0$, the complex number is said to be purely imaginary.
- If $y = 0$, the complex number is said to be real. Zero is the only number which is at once real and purely imaginary.
- The standard form $x + iy$ is denoted z , with $x = \operatorname{Re}(z)$ and $y = \operatorname{Im}(z)$.
- For instance, $\operatorname{Re}(5 - i7) = 5$ and $\operatorname{Im}(5 - i7) = -7$.
- Numbers of the form $\alpha + i\beta$, $\beta \neq 0$ are called imaginary (non-real complex) numbers.

2.2.1 Rectangular Form (contd.)

Equality of complex numbers

Definition 2.2

Two complex numbers $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ are said to be equal if and only if

$\operatorname{Re}(z_1) = \operatorname{Re}(z_2)$ and $\operatorname{Im}(z_1) = \operatorname{Im}(z_2)$.

That is, $x_1 = x_2$ and $y_1 = y_2$.

- For instance, if $\alpha + i\beta = -7 + 3i$, then $\alpha = -7$ and $\beta = 3$.

2.2.2 Argand Plane

Representing complex numbers geometrically

- A complex number $z = x + iy$ is uniquely determined by an ordered pair of real numbers (x, y) .
- The numbers $3 - 8i$, 6 , and $-4i$ are equivalent to $(3, -8)$, $(6, 0)$, and $(0, -4)$ respectively.
- If we consider the x-axis as the real axis and the y-axis as the imaginary axis, the xy-plane is called the complex plane or Argand plane.
- Named after the Swiss mathematician Jean Argand (1768 – 1822).
- A complex number is represented not only by a point, but also by a position vector pointing from the origin to that point.
- $\mathbb{C}$ denotes the set of all complex numbers. Geometrically, a complex number can be viewed as either a point in $\mathbb{R}^2$ or a vector in the Argand plane.

Illustration 2.1

Some complex numbers plotted in the Argand plane

$$2+i, -1+2i, 3-2i, 0-2i, \\ 3+\sqrt{-2}, -2-3i, \cos(\pi/6)+i \sin(\pi/6), 3+0i$$

These are examples of complex numbers written in rectangular form.
Some of them can be plotted as points or vectors in the Argand plane.

2.2.3 Algebraic Operations — (i) Scalar Multiplication

Multiplying a complex number by a real scalar

$$\text{If } z = x + iy \text{ and } k \in \mathbb{R}, \text{ then } kz = (kx) + (ky)i$$

In particular: $0z = 0$, $1z = z$, $(-1)z = -z$

Geometrically, kz stretches (or shrinks) and possibly reverses the direction of the vector z , without changing the line through the origin on which it lies.

(ii) Addition of Complex Numbers

If $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$, where $x_1, x_2, y_1, y_2 \in \mathbb{R}$

$$z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$$

By the parallelogram law of addition, the sum $z_1 + z_2$ corresponds to the point $(x_1 + x_2, y_1 + y_2)$.

The points z_1 , z_2 , and $z_1 + z_2$ in the complex plane may be obtained vectorially — $z_1 + z_2$ is the diagonal of the parallelogram formed by z_1 and z_2 .

(iii) Subtraction of Complex Numbers

The difference $z_1 - z_2$ as a position vector

$$z_1 - z_2 = z_1 + (-z_2) = (x_1 - x_2) + i(y_1 - y_2)$$

The difference $z_1 - z_2$ can be drawn as a position vector whose initial point is the origin and terminal point is $(x_1 - x_2, y_1 - y_2)$.

It may also be drawn joining the endpoint of z_2 to the tip of z_1 — this does not change the meaning of the difference.

(iv) Multiplication of Complex Numbers

The product of two complex numbers z_1 and z_2

$$\begin{aligned} z_1 z_2 &= (x_1 + iy_1)(x_2 + iy_2) \\ &= (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + x_2 y_1) \end{aligned}$$

Although $z_1 z_2$ is itself a complex number represented by a vector, that vector lies in the same plane as z_1 and z_2 .

This product is neither the scalar product nor the vector product used in vector algebra.

Remark — Multiplication of z by i

Multiplying a complex number by i rotates it

$$\text{If } z = x + iy, \text{ then } iz = i(x+iy) = -y + ix$$

The complex number iz is a rotation of z by 90° ($\pi/2$ radians) in the counter-clockwise direction about the origin.

In general, multiplying z by i successively gives a 90° counter-clockwise rotation successively about the origin.

Illustration 2.2

Worked examples of addition, subtraction and multiplication

Let $z_1 = 6+7i$ and $z_2 = 3-5i$. Then z_1+z_2 and z_1-z_2 are:

- $(3-5i)+(6+7i) = (3+6)+(-5+7)i = 9+2i$
- $(6+7i)-(3-5i) = (6-3)+(7-(-5))i = 3+12i$

Let $z_1 = 2+3i$ and $z_2 = 4+7i$. Then z_1z_2 is:

- $(2+3i)(4+7i) = (2 \times 4 - 3 \times 7) + i(2 \times 7 + 3 \times 4)$
- $= (8-21) + (14+12)i = -13+26i$